

Applied NLP

DATA 641

Lecture 14 — Non-negative matrix factorization for topic modeling

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Topic Modeling via Non-Negative Matrix Factorization (NMF)

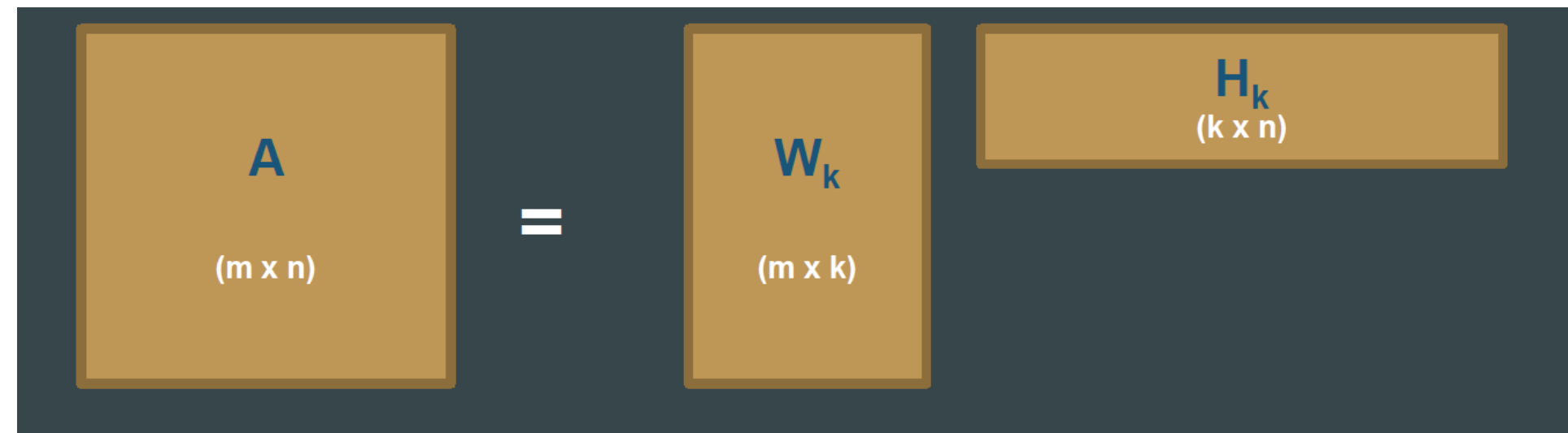
Imagine you have a matrix of data, perhaps representing the relationships between documents and terms. NMF acts as this illuminating spotlight, breaking the data down into two matrices:

- **W** (Basis Vectors): Think of this as a set of building blocks or basis vectors. Each column of **W** represents a basis vector, which is a positive combination of terms. It's like finding atomic concepts in your data, such as "sports," "science," or "politics."
- **H** (Encoding Coefficients): This matrix provides coefficients for combining the basis vectors to reconstruct the original data. It's like a recipe book that tells you how to combine the atomic concepts (basis vectors) to create each document. For example, it might say, "Document A is 30% sports, 50% science, and 20% politics."

The magic of NMF lies in the fact that it enforces positivity in both **W** and **H**, meaning they only contain non-negative values. This positivity constraint makes NMF particularly suitable for tasks where you want to find additive, non-negative components that represent underlying patterns or topics in your data.

Latent Semantic Analysis via NMF

- Use SVD to come up with a rank— r approximation to the matrix $\mathbf{A}$:


$$\begin{matrix} \mathbf{A} \\ (m \times n) \end{matrix} = \begin{matrix} \mathbf{W}_k \\ (m \times k) \end{matrix} \begin{matrix} \mathbf{H}_k \\ (k \times n) \end{matrix}$$

- $\mathbf{W}_k$ assigns m terms to k topics.
- $\mathbf{H}_k$ contains topic distribution per each of the n documents

Underlying mechanism of NMF

NMF iteratively optimizes to factorize the original matrix $\mathbf{A}$ into matrices $\mathbf{W}$ and $\mathbf{H}$ while minimizing the approximation error.

- **Initialization:** Initialize $\mathbf{W}$ and $\mathbf{H}$ with non-negative random values or using some other initialization method.
- **Update $\mathbf{W}$ and $\mathbf{H}$ Alternately:** NMF uses an iterative approach to update $\mathbf{W}$ and $\mathbf{H}$ alternately until convergence. The typical update steps are as follows:
 - **Update $\mathbf{W}$:** Fix $\mathbf{H}$ and update $\mathbf{W}$ by minimizing the approximation error between $\mathbf{A}$ and $\mathbf{WH}$ while enforcing non-negativity. This update is often performed using optimization techniques like gradient descent.
 - **Update $\mathbf{H}$:** Fix $\mathbf{W}$ and update $\mathbf{H}$ in a similar way by minimizing the error between $\mathbf{A}$ and $\mathbf{WH}$ while keeping the values non-negative.
- **Convergence Criterion:** Check a convergence criterion, such as the change in the approximation error or the number of iterations. If the criterion is met, stop the iteration; otherwise, continue with the updates.
- **Resulting Matrices $\mathbf{W}$ and $\mathbf{H}$:** Upon convergence, $\mathbf{W}$ embodies basis vectors (topics), while $\mathbf{H}$ contains encoding coefficients for reconstructing $\mathbf{A}$.

Advantages of NMF for Topic Modeling

- **Non-Negativity:** NMF enforces non-negativity in both the basis vectors and the encoding coefficients. This constraint makes the resulting topics and document-topic distributions more interpretable, as negative values are often challenging to explain in the context of topics.
- **Parts-Based Representation:** NMF provides a parts-based representation of data, which means it can capture additive components that resemble meaningful topics or patterns. This makes NMF particularly suitable for cases where topics are composed of subtopics or atomic concepts.
- **Interpretability:** NMF tends to generate topics that are more human-interpretable than some other techniques. This makes it easier for users to understand and label topics, which is crucial in applications like content recommendation and information retrieval.
- **Sparse Solutions:** NMF often produces sparse representations, meaning that only a subset of basis vectors is used to represent each document. This can help reduce dimensionality and improve efficiency in topic modeling.
- **Applicability to Various Data Types:** NMF is versatile and can be applied to various types of data, including text, images, and gene expression data. This makes it useful in a wide range of domains.

Disadvantages of SVD for Topic Modeling

- **Lack of Orthogonality:** Unlike Principal Component Analysis (PCA), the basis vectors obtained from NMF are not guaranteed to be orthogonal. This can lead to some overlap between topics, making them less distinct.
- **Determining the Number of Topics:** Like many topic modeling methods, NMF requires you to specify the number of topics in advance, which can be challenging in practice. Selecting an inappropriate number of topics can affect the quality of results.
- **Initialization Sensitivity:** NMF's performance can be sensitive to the choice of initialization. Different initializations may lead to different results, making it necessary to run the algorithm multiple times with different starting points.
- **Computational Complexity:** NMF can be computationally demanding, especially for large datasets or when using high-dimensional feature spaces.
- **Local Optima:** NMF optimization can get stuck in local optima. Ensuring convergence to a good solution may require additional strategies, such as random restarts.
- **Limited Negative Modeling:** Since NMF enforces non-negativity, it may not capture negative correlations between terms, which could be relevant in some contexts.